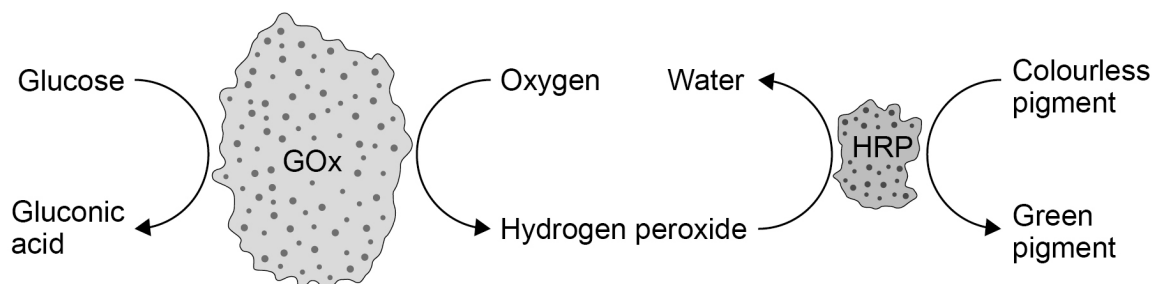


0 8

A scientist investigated a sequence of reactions catalysed by **two** enzymes, GOx and HRP. **Figure 9** shows this sequence of reactions.

Figure 9



0 8 . 1

Use **Figure 9** to identify all of the products formed when this sequence of reactions is completed.

[1 mark]

0 8 . 2

The scientist joined DNA molecules together to make tiny cages. The cages are exactly 20 nm long, 20 nm wide and 17 nm deep.

He trapped **one** GOx molecule and **one** HRP molecule together in each cage. The GOx molecule and HRP molecule fill 9% of the cage volume.

The volume of a GOx molecule is eight times larger than an HRP molecule.

Use this information to calculate the volume of a GOx molecule. Give the appropriate unit with your answer.

Show your working.

[3 marks]

Answer _____



Question 8 continues on the next page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



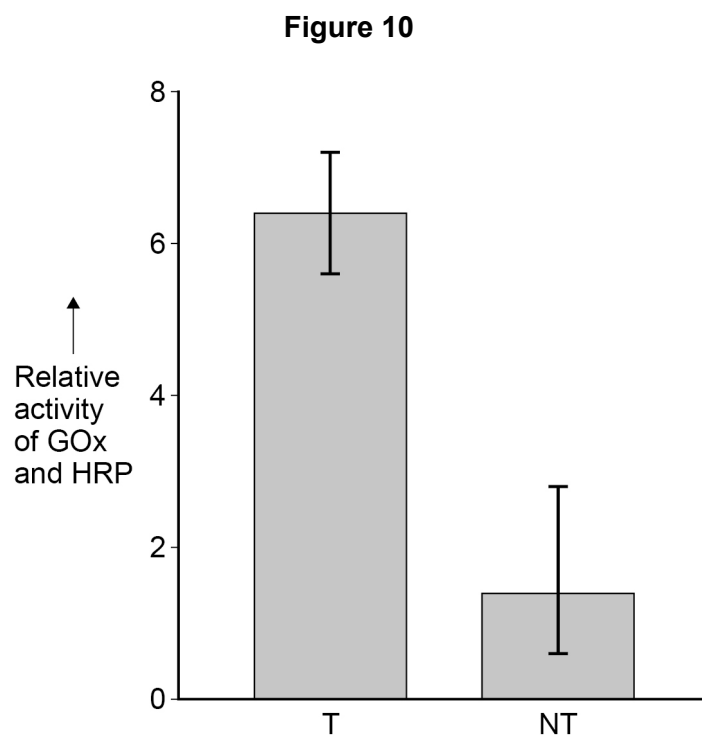
The scientist investigated the activity of GOx and HRP enzymes when they are:

- trapped inside cages (**T**) and
- not trapped (**NT**), but free in solution with **no** cages.

Figure 10 shows his results.

The error bars show ± 2 standard deviations.

± 2 standard deviations include 95% of the data.



0 8 . 3

What can you conclude from **Figure 10** about the effect of trapping GOx and HRP inside cages?

[3 marks]

0 8 . 4

The design of the scientist's investigation did **not** include a suitable control.

Suggest a suitable control.

[1 mark]

8

Turn over for the next question

Turn over ►